

Lab 13: CDC → Kafka → Downstream Consumer with a Resolved-Timestamp Frontier (55 min)

Learning Objectives

By the end of this lab you will be able to:

- Run a Core changefeed and an Enterprise changefeed and explain when each is appropriate
- Emit to Kafka with `updated`, `resolved`, `diff`, and `key_in_value` options and read the envelopes
- Track the **resolved-timestamp frontier** in a consumer and know when data is safe to act on
- Build an **idempotent consumer** that survives at-least-once delivery and replays
- Measure changefeed lag with `changefeed_max_behind_nanos` and know what makes it grow
- Handle schema changes under an active changefeed without breaking downstream consumers
- Pause, resume, and restart a changefeed from a cursor after a downstream outage

Prerequisites

- **Docker Desktop** (or Docker Engine) running — there is no `cockroach` binary to install
- Docker (for Kafka). A webhook-sink fallback with `nc` is provided if Docker is unavailable.
- `python3` for the consumer

Setup

1. Cluster

```
scripts/crdb up
export C='postgresql://root@localhost:26257?sslmode=disable'
```

The cluster runs in Docker (see [Lab 1](#)). From your machine it is `localhost:26257`; from inside another container it is `crdb1:26257`. `scripts/crdb run ...` executes inside node 1.

Enterprise changefeeds need a licence — the class has one. It ships with the repo in `.license.env`: `scripts/crdb up` prints `enterprise licence applied` when it starts, and nothing further is needed. If your cluster predates that file, `scripts/crdb reset`. Without a licence, `CREATE CHANGEFEED ... INTO` is refused and you run Part A only — core changefeeds, which need no licence and teach the same envelope format.

```
scripts/crdb sql <<'SQL'
SET CLUSTER SETTING kv.rangefeed.enabled = true;

CREATE DATABASE shop;
USE shop;

CREATE TABLE orders (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  customer    STRING NOT NULL,
  total       DECIMAL(12,2) NOT NULL,
  status      STRING NOT NULL DEFAULT 'new',
  updated     TIMESTAMPTZ NOT NULL DEFAULT now()
);
SQL
```

2. Kafka

```
mkdir -p /tmp/lab13

cat > /tmp/lab13/docker-compose.yml <<'YML'
services:
  kafka:
    image: apache/kafka:3.9.0
    container_name: kafka
    hostname: kafka
    # Kafka joins the lab cluster's network, because the *database* is what
    # connects to it. A changefeed sink is dialled by the CockroachDB node, not
    # by your shell.
    networks: [crdb]
    environment:
      KAFKA_NODE_ID: "0"
      KAFKA_PROCESS_ROLES: "controller,broker"
      KAFKA_LISTENERS: "PLAINTEXT://:9092,CONTROLLER://:9093"
      KAFKA_ADVERTISED_LISTENERS: "PLAINTEXT://kafka:9092"
      KAFKA_CONTROLLER_QUORUM_VOTERS: "0@kafka:9093"
      KAFKA_CONTROLLER_LISTENER_NAMES: "CONTROLLER"
      KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: "CONTROLLER:PLAINTEXT,PLAINTEXT:PLAINTEXT"
      KAFKA_INTER_BROKER_LISTENER_NAME: "PLAINTEXT"
      KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: "1"
      KAFKA_TRANSACTION_STATE_LOG_REPLICATION_FACTOR: "1"
      KAFKA_TRANSACTION_STATE_LOG_MIN_ISR: "1"
      KAFKA_AUTO_CREATE_TOPICS_ENABLE: "true"

networks:
  crdb:
    external: true
    name: crdb-labs_default
YML

docker compose -f /tmp/lab13/docker-compose.yml up -d
until docker compose -f /tmp/lab13/docker-compose.yml logs kafka 2>/dev/null | grep -q
  "Kafka Server started"; do sleep 2; done; echo "Kafka is up"
```

The official `apache/kafka` image, not `bitnami/kafka`. Bitnami withdrew its versioned Docker Hub tags, so `bitnami/kafka:3.7` now fails to pull. The Apache image drops the `_CFG` infix from every environment variable and keeps its CLI tools in `/opt/kafka/bin` rather than on `PATH` — which is why the commands below spell out the full path.

⚠ **The cluster must already be up** (step 1) — this compose file joins the network that stack creates, and refuses to start if it does not exist yet. Stay in the repository root: every command in this lab, including `docker compose -f /tmp/lab13/docker-compose.yml ...`, is written to run from there.

Why `kafka:9092` and not `localhost:9092`? The changefeed is opened *by the database node*, inside its own container. `localhost` there is the node itself, not your machine, so a sink of `kafka://localhost:9092` fails with a connection refused that looks baffling until you remember whose `localhost` it is. Advertising `kafka:9092` on the shared network is what makes the sink reachable from where it is actually dialled. Every `kafka-console-consumer` command below runs *inside* the Kafka container (`docker compose -f /tmp/lab13/docker-compose.yml exec kafka ...`), which is why those still say `localhost:9092`.

Tasks

Part A: Core Changefeed — the Zero-Infrastructure Version (8 min)

A Core changefeed streams straight to the SQL session. No sink, no license, no delivery guarantees beyond "this session is connected".

1. Terminal A:

```
scripts/crdb sql -e "EXPERIMENTAL CHANGEFEED FOR shop.orders WITH updated;"
```

2. Terminal B:

```
scripts/crdb sql -e "INSERT INTO shop.orders (customer, total) VALUES ('alice', 42.00);"  
scripts/crdb sql -e "UPDATE shop.orders SET status = 'paid' WHERE customer = 'alice';"  
scripts/crdb sql -e "DELETE FROM shop.orders WHERE customer = 'alice';"
```

Three commands, not one `-e` string: statements passed together run as one implicit transaction, and a changefeed emits one event per key per transaction — insert-then-delete in a single transaction emits a lone tombstone whose `before` and `after` are both `null`, and the insert and update are never seen (verified).

3. Read the envelopes in terminal A. Note that a delete emits `after: null` — a tombstone.

If you redirect a core changefeed to a file or a pipe, you get nothing. The default `--format=table` buffers all output to compute column widths, so a stream that never ends never prints. Use a streaming format when the output is not a terminal:

```
stdbuf -oL scripts/crdb sql --format=csv \  
-e "EXPERIMENTAL CHANGEFEED FOR shop.orders WITH updated;" > /tmp/cdc.csv
```

Both parts matter. `--format=csv` avoids the column-sizing buffer, and `stdbuf -oL` line-buffers stdout — without it, nothing reaches the file until the process exits, and a changefeed never exits. (On macOS, `stdbuf` comes from `brew install coreutils` as `gstdbuf`; in a terminal you don't need either.) The `key` and `value` columns come back **hex-encoded** (`\x7b2261...`) because they are `BYTES`. Decode them with:

```
python3 -c "import csv,sys  
for r in csv.reader(open('/tmp/cdc.csv')):  
    for c in r:  
        if c.startswith(r'\x'): print(bytes.fromhex(c[2:]).decode('utf-8','replace'))"
```

This is a `cockroach sql` presentation detail, not a changefeed one — an Enterprise changefeed to a real sink emits plain JSON.

Use Core when	Use Enterprise when
Ad-hoc debugging	Anything in production
You need no durability	You need at-least-once delivery
Single consumer, short-lived	Multiple consumers, restartable

Part B: Enterprise Changefeed to Kafka (12 min)

1. Create the changefeed:

```
CREATE CHANGEFEED FOR TABLE shop.orders  
INTO 'kafka://kafka:9092'  
WITH updated, resolved = '10s', diff, key_in_value,  
min_checkpoint_frequency = '10s';
```

Note the returned `job_id`.

2. Confirm the topic exists:

```
docker compose -f /tmp/lab13/docker-compose.yml exec kafka /opt/kafka/bin/kafka-topics.sh  
--bootstrap-server localhost:9092 --list
```

The topic is named after the table: `orders`.

3. Generate changes:

```
scripts/crdb sql <<'SQL'
USE shop;
INSERT INTO orders (customer, total) SELECT 'cust-' || g, (g * 3.50)::DECIMAL(12,2)
FROM generate_series(1, 20) g;
UPDATE orders SET status = 'paid', updated = now() WHERE total > 40;
DELETE FROM orders WHERE total < 10;
SQL
```

4. Consume raw:

```
docker compose -f /tmp/lab13/docker-compose.yml exec kafka /opt/kafka/bin/kafka-console-
consumer.sh \
  --bootstrap-server localhost:9092 --topic orders --from-beginning
```

Ctrl+C when you have seen enough. A changefeed with `resolved` emits a watermark every few seconds, so the stream never goes quiet — which is why `--timeout-ms` (an *inactivity* timeout) would never fire here.

5. Decode an envelope. With `updated`, `diff`, `key_in_value` you get:

```
{
  "after": {
    "id": "...", "customer": "cust-3", "total": 10.50, "status": "paid", "updated": "...",
  },
  "before": {
    "id": "...", "customer": "cust-3", "total": 10.50, "status": "new", "updated": "...",
  },
  "key": ["..."],
  "updated": "1716400000000000000.0000000000"
}
```

Option	Adds	Costs
<code>updated</code>	MVCC timestamp of the change	negligible
<code>resolved = 'Ns'</code>	Periodic watermark messages	more messages
<code>diff</code>	The <code>before</code> image	~2× payload size
<code>key_in_value</code>	Primary key inside the value	small
<code>format = avro</code>	Avro + schema registry	needs a registry
<code>envelope = 'wrapped'</code> (default)	before/after wrapper	—

6. Watch the job:

```
SELECT job_id, status, running_status, high_water_timestamp
FROM [SHOW CHANGEFEED JOBS];
```

`high_water_timestamp` is the frontier: every change with an MVCC timestamp at or below it has already been emitted.

Part C: The Resolved-Timestamp Frontier (15 min)

This is the part everyone gets wrong. Changefeed messages arrive **out of order across ranges**. A `resolved` message is the cluster telling you: *no further message will arrive with a timestamp below this value*. Only then is a time window complete.

1. Watch the resolved messages arrive:

```
docker compose -f /tmp/lab13/docker-compose.yml exec kafka /opt/kafka/bin/kafka-console-  
consumer.sh \  
  --bootstrap-server localhost:9092 --topic orders \  
  | grep --line-buffered resolved          # Ctrl+C to stop
```

2. Build a frontier-tracking consumer — `/tmp/lab13/consumer.py`:

```
#!/usr/bin/env python3
"""Frontier-tracking, idempotent CDC consumer.

Buffers rows until the resolved timestamp passes them, then applies them
exactly once (keyed by primary key + MVCC timestamp).
"""
import json, sys, subprocess
from collections import defaultdict

frontier = "0"           # highest resolved timestamp seen
pending  = []            # (updated_ts, key, payload) not yet final
applied  = {}            # key -> last applied timestamp (idempotency ledger)

def apply(key, ts, payload):
    # At-least-once delivery means we WILL see duplicates and replays.
    # Applying only when ts > last-applied makes the consumer idempotent.
    if key in applied and ts <= applied[key]:
        print(f" skip duplicate {key} @ {ts}")
        return
    applied[key] = ts
    action = "DELETE" if payload.get("after") is None else "UPSERT"
    print(f" {action:6} {key} @ {ts}")

def handle(line):
    global frontier
    try:
        msg = json.loads(line)
    except json.JSONDecodeError:
        return
    if "resolved" in msg:
        frontier = msg["resolved"]
        ready = [p for p in pending if p[0] <= frontier]
        for ts, key, payload in sorted(ready):
            apply(key, ts, payload)
        pending[:] = [p for p in pending if p[0] > frontier]
        print(f"FRONTIER advanced to {frontier} "
              f"({len(ready)} applied, {len(pending)} still pending)")
        return
    key = json.dumps(msg.get("key"))
    ts = msg.get("updated", "0")
    pending.append((ts, key, msg))

for line in sys.stdin:
    handle(line.strip())
```

3. Run it against the live topic:

```
docker compose -f /tmp/lab13/docker-compose.yml exec -T kafka /opt/kafka/bin/kafka-
  console-consumer.sh \
  --bootstrap-server localhost:9092 --topic orders --from-beginning \
  | python3 -u /tmp/lab13/consumer.py
```

Leave this running — it prints each `FRONTIER advanced` as it happens. Ctrl+C when done.

4. In another terminal, make changes and watch the frontier advance:

```
scripts/crdb sql -d shop -e
  "INSERT INTO orders (customer, total) VALUES ('frontier-test', 99.99);"
scripts/crdb sql -d shop -e "UPDATE orders SET status = 'shipped' WHERE customer =
  'frontier-test';"
```

Two separate commands on purpose: statements passed together in one `-e` string run as a single implicit transaction, and a changefeed emits one event per key per transaction — you would see a single `UPSERT` with `status: shipped` instead of an insert followed by an update. The rows sit in `pending` until the next `resolved` message, then apply in timestamp order.

5. The three questions the frontier answers:

Question	Answer
"Is my materialized view up to date as of 14:00?"	Yes, iff frontier ≥ 14:00
"Can I close the books for yesterday?"	Yes, iff frontier ≥ midnight
"Did I miss any events during the outage?"	No, iff you restart from the last frontier

6. Tune the trade-off. Lower `resolved` interval = fresher frontier = more messages:

```
-- Recreate with a 1-second frontier and compare message volume
CREATE CHANGEFEED FOR TABLE shop.orders INTO 'kafka://kafka:9092?topic_prefix=fast_'
  WITH updated, resolved = '1s';
```

Part D: Idempotency and Replay (10 min)

Changefeeds guarantee **at-least-once** delivery, not exactly-once. Duplicates are normal: after a job restart, a lease transfer, or a rebalance, messages are re-emitted.

1. Force a replay — restart the changefeed from an earlier cursor:

```
SELECT job_id, high_water_timestamp FROM [SHOW CHANGEFEED JOBS];
PAUSE JOB <job_id>;

CREATE CHANGEFEED FOR TABLE shop.orders
  INTO 'kafka://kafka:9092?topic_prefix=replay_'
  WITH updated, resolved = '10s', cursor = '<a timestamp from 5 minutes ago>';
```

2. Run the consumer against the replay topic and confirm your `applied` ledger skips the duplicates it already has.

3. The three ways to be idempotent downstream:

Technique	How	Best for
Upsert by primary key	<code>INSERT ... ON CONFLICT DO UPDATE</code>	Materialized copies
Timestamp guard	Apply only if <code>updated > last_seen</code>	Out-of-order safety
Dedup ledger	Store <code>(key, ts)</code> seen-set with TTL	Side effects (emails, charges)

For side effects, the ledger is the only safe option — you cannot "re-send" an email idempotently by upserting a row.

4. **Verify no data was lost.** Compare the source with what your consumer applied:

```
SELECT count(*) FROM shop.orders;
```

Against the consumer's `applied` map size, minus the deletes.

Part E: Schema Changes Under an Active Changefeed (10 min)

1. **With the changefeed running, add a column:**

```
ALTER TABLE shop.orders ADD COLUMN shipped_at TIMESTAMPTZ;  
INSERT INTO shop.orders (customer, total, shipped_at) VALUES ('post-alter', 12.00, now());
```

2. **Watch the consumer.** The envelope now carries the new field. A consumer that parses strictly (fixed field list) breaks; one that reads by name survives.

3. **Control the behaviour explicitly:**

```
CREATE CHANGEFEED FOR TABLE shop.orders INTO 'kafka://kafka:9092?topic_prefix=strict_'  
WITH updated, resolved='10s', schema_change_policy = 'stop';
```

schema_change_policy	Behaviour
backfill (default)	Emit a full backfill of the table after the change
nobackfill	Continue without re-emitting existing rows
stop	Fail the job so a human decides

4. **The rule for downstream contracts:** additive changes (new nullable column) are safe; renames and type changes are not. Version your consumer's schema handling the same way you'd version an API.

Optional — If Time Allows

These parts are not required to complete the lab; they extend it by about 10 minutes. Do them if you finish early, or after class — the cluster and data from the core parts carry over.

Part F: Lag, Failure & Recovery (10 min)

1. **Measure lag:**

```
-- high_water_timestamp is an HLC DECIMAL; hlc_to_timestamp() converts it.
-- (A direct ::DECIMAL::TIMESTAMPTZ cast is rejected: 'invalid cast'.)
SELECT job_id,
       (now() - hlc_to_timestamp(high_water_timestamp)) AS lag
FROM [SHOW CHANGEFEED JOBS] WHERE status = 'running';
```

```
curl -s http://localhost:8080/_status/vars | grep -E 'changefeed_(max_behind_nanos|
checkpoint_progress)' | grep -v '^#'
```

2. Kill the sink and watch the job react:

```
docker compose -f /tmp/lab13/docker-compose.yml stop kafka
```

```
SELECT job_id, status, running_status FROM [SHOW CHANGEFEED JOBS];
```

The job retries; it does not silently drop events. `protected timestamps` keep the underlying MVCC data from being garbage-collected while the feed is behind.

3. Bring the sink back:

```
docker compose -f /tmp/lab13/docker-compose.yml start kafka
```

```
SELECT job_id, status, high_water_timestamp FROM [SHOW CHANGEFEED JOBS];
```

The frontier resumes from where it stopped. Nothing was lost.

The hidden cost: while a changefeed is paused or behind, its protected timestamp prevents GC of old MVCC versions. A changefeed left paused for days grows your storage and slows scans. `SHOW JOBS` is where you find the paused feed nobody remembered.

4. Cancel cleanly:

```
CANCEL JOB <job_id>;
SELECT job_id, status FROM [SHOW CHANGEFEED JOBS];
```

Cleanup

```
docker compose -f /tmp/lab13/docker-compose.yml down -v
scripts/crdb down
```

Lab 13 Deliverables

□ **Core and Enterprise changefeeds** both run, with a stated rule for when to use each □ **Kafka sink** with `updated`, `resolved`, `diff`, `key_in_value` and the envelopes decoded □ **Frontier consumer** that buffers until resolved and applies in timestamp order □ **Idempotency** proven by replaying from an earlier cursor □ **Schema change** handled under an active feed, with the policy options understood □ **Failure drill**: sink killed and restored with no data loss, lag measured

Challenge Exercises

1. **Avro + schema registry.** Add Confluent Schema Registry to the compose file and re-create the changefeed with `format = avro`, `confluent_schema_registry = '...'`. What does the consumer gain?
2. **Fan out by tenant.** Use `WITH topic_in_value` and a changefeed on a `REGIONAL BY ROW` table. Can you route each region's events to a different topic?
3. **Exactly-once, end to end.** Combine the dedup ledger with Kafka transactions in your consumer. Where is the remaining window in which a duplicate side effect could occur?
4. **Alert on lag.** Add a Prometheus alert (Lab 9) on `changefeed_max_behind_nanos > 60e9` and prove it fires by pausing the job.

Reference

Command	Purpose
<code>EXPERIMENTAL CHANGEFEED FOR t</code>	Core changefeed to the SQL session
<code>CREATE CHANGEFEED FOR TABLE t INTO 'kafka://...'</code>	Enterprise changefeed
<code>WITH resolved = '10s'</code>	Emit frontier watermarks
<code>WITH diff</code>	Include the <code>before</code> image
<code>WITH cursor = '<ts>'</code>	Start (or replay) from a timestamp
<code>WITH schema_change_policy = 'stop'</code>	Fail on schema change instead of backfilling
<code>SHOW CHANGEFEED JOBS</code>	Status and <code>high_water_timestamp</code>
<code>PAUSE / RESUME / CANCEL JOB</code>	Feed lifecycle
<code>changefeed_max_behind_nanos</code>	Lag metric for alerting